Checkpoint Engine and Online Weight Publication

Deep Dive into Kimi K3 — Chapter 41

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Goal: separate received bytes, installed shards, admitted requests, and fleet activation — and keep each guarantee, measurement, and CE attribution in scope.

Roadmap

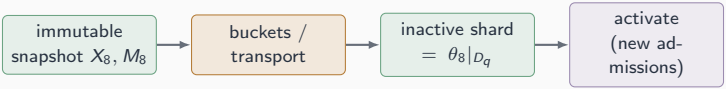
A coherent version, exactly placed

Moving and installing bytes

Activation, recovery, and evidence

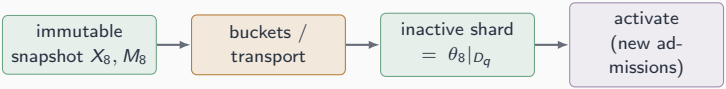
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- A coherent snapshot decodes to one training update; per-tensor checksums cannot detect a W_7 mixed with b_8.
- Row-owned trainers feed column-owned workers through $S_r \cap D_q$: coverage, authority, and placement give exact reconstruction. 28 scalars, 112 logical bytes, 128 bytes per replica.

[DERIVED: chapter fixture]

Moving and installing bytes

Checkpoint Engine path, receivers, and overlap

- CE: parameter server beside inference workers; two alternating IPC slots whose reuse is justified by completion; release and postprocessing are separate gates. [INSPECTED: Moonshot CE source]

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- Ideal pipeline $C_m = a + b + (m-1)\max(a, b)$: 14 vs. 20 synthetic units — illegal if stages share one engine. Shadow copy peaks at 288 bytes, over a 272-byte budget.

Activation, recovery, and evidence

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Publication supplies usable identity; the scheduler decides how to use it. Next: Chapter 42's Million-Token Agentic-RL Infrastructure.